

Part II

Implementing Ethernet LANs

Chapter 4: Using the Command-Line Interface

Chapter 5: Analyzing Ethernet LAN Switching

Chapter 6: Configuring Basic Switch Management

Chapter 7: Configuring and Verifying Switch Interfaces

Part II Review



CHAPTER 4

Using the Command-Line Interface

This chapter covers the following exam topics:

None

This chapter explains foundational skills required before you can learn about the roughly 15 exam topics that use the verbs *configure* and *verify*. However, Cisco does not list the foundational skills described in this chapter as a separate exam topic, so there are no specific exam topics included in this chapter.

To create an Ethernet LAN, network engineers start by planning. They consider the requirements, create a design, buy the switches, contract to install cables, and configure the switches to use the right features.

The CCNA exam focuses on skills like understanding how LANs work, configuring different switch features, verifying that those features work correctly, and finding the root cause of the problem when a feature is not working correctly. The first skill you need to learn before doing all the configuration and verification tasks is to learn how to access and use the user interface of the switch, called the command-line interface (CLI).

This chapter begins that process by showing the basics of how to access the switch's CLI. These skills include how to access the CLI and how to issue verification commands to check on the status of the LAN. This chapter also includes the processes of how to configure the switch and how to save that configuration.

Note that this chapter focuses on processes that provide a foundation for most every exam topic that includes the verbs *configure* and/or *verify*. Most of the rest of the chapters in Parts II and III of this book then go on to include details of the particular commands you can use to verify and configure different switch features.

“Do I Know This Already?” Quiz

Take the quiz (either here or use the PTP software) if you want to use the score to help you decide how much time to spend on this chapter. The letter answers are listed at the bottom of the page following the quiz. Appendix C, found both at the end of the book as well as on the companion website, includes both the answers and explanations. You can also find both answers and explanations in the PTP testing software.

Table 4-1 “Do I Know This Already?” Foundation Topics Section-to-Question Mapping

Foundation Topics Section	Questions
Accessing the Cisco Catalyst Switch CLI	1–3
Configuring Cisco IOS Software	4–6

1. In what modes can you type the command **show mac address-table** and expect to get a response with MAC table entries? (Choose two answers.)
 - a. User mode
 - b. Enable mode
 - c. Global configuration mode
 - d. Interface configuration mode
2. In which of the following modes of the CLI could you type the command **reload** and expect the switch to reboot?
 - a. User mode
 - b. Enable mode
 - c. Global configuration mode
 - d. Interface configuration mode
3. Which of the following is a difference between Telnet and SSH as supported by a Cisco switch?
 - a. SSH encrypts the passwords used at login, but not other traffic; Telnet encrypts nothing.
 - b. SSH encrypts all data exchange, including login passwords; Telnet encrypts nothing.
 - c. Telnet is used from Microsoft operating systems, and SSH is used from UNIX and Linux operating systems.
 - d. Telnet encrypts only password exchanges; SSH encrypts all data exchanges.
4. What type of switch memory is used to store the configuration used by the switch when it is up and working?
 - a. RAM
 - b. ROM
 - c. Flash
 - d. NVRAM
 - e. Bubble
5. What command copies the configuration from RAM into NVRAM?
 - a. **copy running-config tftp**
 - b. **copy tftp running-config**
 - c. **copy running-config start-up-config**
 - d. **copy start-up-config running-config**
 - e. **copy startup-config running-config**
 - f. **copy running-config startup-config**

6. A switch user is currently in console line configuration mode. Which of the following would place the user in enable mode? (Choose two answers.)
- a. Using the **exit** command once
 - b. Using the **end** command once
 - c. Pressing the Ctrl+Z key sequence once
 - d. Using the **quit** command

Foundation Topics

Accessing the Cisco Catalyst Switch CLI

Cisco uses the concept of a command-line interface (CLI) with its router products and most of its Catalyst LAN switch products. The CLI is a text-based interface in which the user, typically a network engineer, enters a text command and presses Enter. Pressing Enter sends the command to the switch, which tells the device to do something. The switch does what the command says, and in some cases, the switch replies with some messages stating the results of the command.

Cisco Catalyst switches also support other methods to both monitor and configure a switch. For example, a switch can provide a web interface so that an engineer can open a web browser to connect to a web server running in the switch. Switches also can be controlled and operated using network management software.

This book discusses only Cisco Catalyst enterprise-class switches, and in particular, how to use the Cisco CLI to monitor and control these switches. This first major section of the chapter first examines these Catalyst switches in more detail and then explains how a network engineer can get access to the CLI to issue commands.

Cisco Catalyst Switches

Within the Cisco Catalyst brand of LAN switches, Cisco produces a wide variety of switch series or families. Each switch series includes several specific models of switches that have similar features, similar price-versus-performance tradeoffs, and similar internal components.

For example, at the time this book was published, the Cisco 2960-XR series of switches was a current switch model series. Cisco positions the 2960-XR series (family) of switches as full-featured, low-cost wiring closet switches for enterprises. That means that you would expect to use 2960-XR switches as access switches in a typical campus LAN design.

Figure 4-1 shows a photo of 10 different models from the 2960-XR switch model series from Cisco. Each switch series includes several models, with a mix of features. For example, some of the switches have 48 RJ-45 unshielded twisted-pair (UTP) 10/100/1000 ports, meaning that these ports can autonegotiate the use of 10BASE-T (10 Mbps), 100BASE-T (100 Mbps), or 1000BASE-T (1 Gbps) Ethernet.



Figure 4-1 Cisco 2960-XR Catalyst Switch Series

Cisco refers to a switch's physical connectors as either *interfaces* or *ports*, with an interface type and interface number. The interface type, as used in commands on the switch, is either Ethernet, Fast Ethernet, Gigabit Ethernet, and so on for faster speeds. For Ethernet interfaces that support running at multiple speeds, the permanent name for the interface refers to the fastest supported speed. For example, a 10/100/1000 interface (that is, an interface that runs at 10 Mbps, 100 Mbps, or 1000 Mbps) would be called Gigabit Ethernet no matter what speed is currently in use.

To uniquely number each different interface, some Catalyst switches use a two-digit interface number (x/y), while others have a three-digit number ($x/y/z$). For instance, two 10/100/1000 ports on many older Cisco Catalyst switches would be called GigabitEthernet 0/0 and GigabitEthernet 0/1, while on the newer 2960-XR series, two interfaces would be GigabitEthernet 1/0/1 and GigabitEthernet 1/0/2.

Accessing the Cisco IOS CLI

Like any other piece of computer hardware, Cisco switches need some kind of operating system software. Cisco calls this OS the Internetwork Operating System (IOS).

Cisco IOS Software for Catalyst switches implements and controls logic and functions performed by a Cisco switch. Besides controlling the switch's performance and behavior, Cisco IOS also defines an interface for humans called the CLI. The Cisco IOS CLI allows the user to use a terminal emulation program, which accepts text entered by the user. When the user presses Enter, the terminal emulator sends that text to the switch. The switch processes the text as if it is a command, does what the command says, and sends text back to the terminal emulator.

The switch CLI can be accessed through three popular methods—the console, Telnet, and Secure Shell (SSH). Two of these methods (Telnet and SSH) use the IP network in which the switch resides to reach the switch. The console is a physical port built specifically to allow access to the CLI. Figure 4-2 depicts the options.

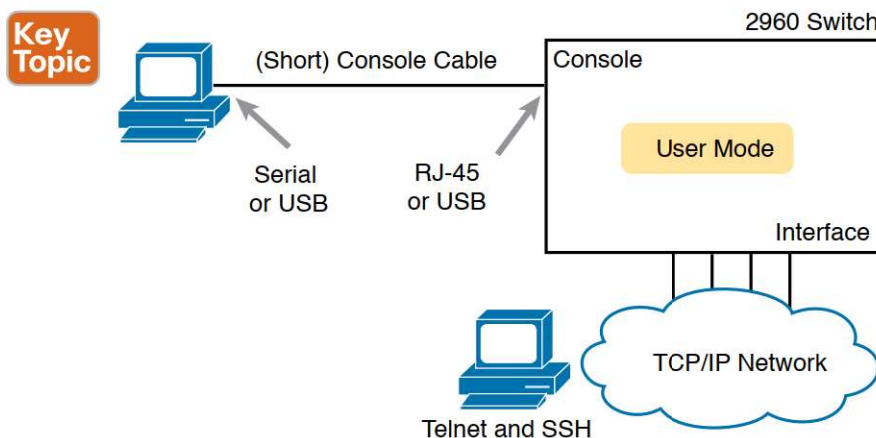


Figure 4-2 CLI Access Options

Console access requires both a physical connection between a PC (or other user device) and the switch's console port, as well as some software on the PC. Telnet and SSH require software on the user's device, but they rely on the existing TCP/IP network to transmit data. The next few pages detail how to connect the console and set up the software for each method to access the CLI.

Cabling the Console Connection

The physical console connection, both old and new, uses three main components: the physical console port on the switch, a physical serial port on the PC, and a cable that works with the console and serial ports. However, the physical cabling details have changed slowly over time, mainly because of advances and changes with serial interfaces on PC hardware. For this next topic, the text looks at three cases: newer connectors on both the PC and the switch, older connectors on both, and a third case with the newer (USB) connector on the PC but with an older connector on the switch.

Most PCs today use a familiar standard USB cable for the console connection. Cisco has been including USB ports as console ports in newer routers and switches as well. All you have to do is look at the switch to make sure you have the correct style of USB cable end to match the USB console port. In the simplest form, you can use any USB port on the PC, with a USB cable, connected to the USB console port on the switch or router, as shown on the far right side of Figure 4-3.

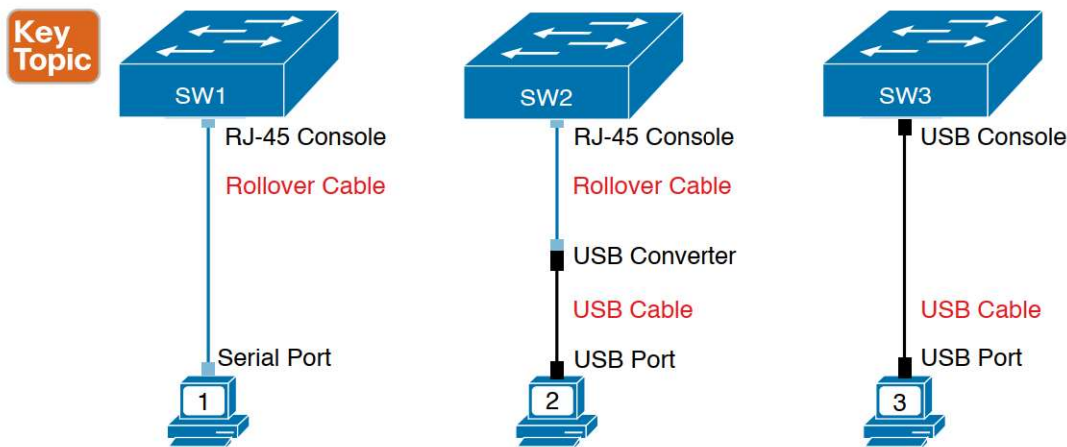


Figure 4-3 Console Connection to a Switch

Older console connections use a PC serial port that pre-dates USB, a UTP cable, and an RJ-45 console port on the switch, as shown on the left side of Figure 4-3. The PC serial port typically has a D-shell connector (roughly rectangular) with nine pins (often called a DB-9). The console port looks like any Ethernet RJ-45 port (but is typically colored in blue and with the word *console* beside it on the switch).

The cabling for this older-style console connection can be simple or require some effort, depending on what cable you use. You can use the purpose-built console cable that ships with new Cisco switches and routers and not think about the details. However, you can make

Answers to the “Do I Know This Already?” quiz:

1 A, B **2** B **3** B **4** A **5** F **6** B, C

your own cable with a standard serial cable (with a connector that matches the PC), a standard RJ-45 to DB-9 converter plug, and a UTP cable. However, the UTP cable does not use the same pinouts as Ethernet; instead, the cable uses rollover cable pinouts rather than any of the standard Ethernet cabling pinouts. The rollover pinout uses eight wires, rolling the wire at pin 1 to pin 8, pin 2 to pin 7, pin 3 to pin 6, and so on.

As it turns out, USB ports became common on PCs before Cisco began commonly using USB for its console ports. So, you also have to be ready to use a PC that has only a USB port and not an old serial port, but a router or switch that has the older RJ-45 console port (and no USB console port). The center of Figure 4-3 shows that case. To connect such a PC to a router or switch console, you need a USB converter that converts from the older console cable to a USB connector, and a rollover UTP cable, as shown in the middle of Figure 4-3.

NOTE When using the USB options, you typically also need to install a software driver so that your PC's OS knows that the device on the other end of the USB connection is the console of a Cisco device. Also, you can easily find photos of these cables and components online, with searches like “cisco console cable,” “cisco usb console cable,” or “console cable converter.”

The 2960-XR series, for instance, supports both the older RJ-45 console port and a USB console port. Figure 4-4 points to the two console ports; you would use only one or the other. Note that the USB console port uses a mini-B port rather than the more commonly seen rectangular standard USB Type A port.

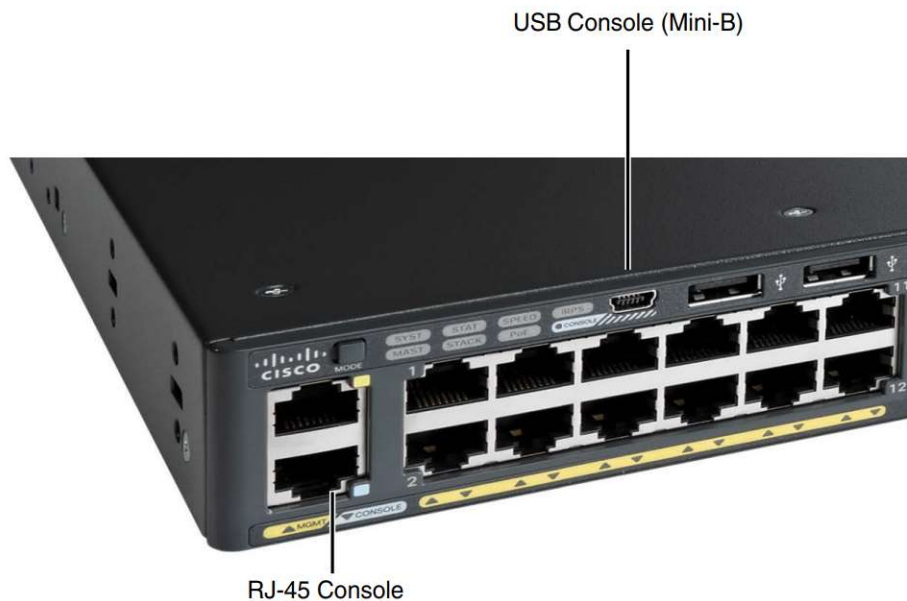


Figure 4-4 A Part of a 2960-XR Switch with Console Ports Shown

After the PC is physically connected to the console port, a terminal emulator software package must be installed and configured on the PC. The terminal emulator software treats all data as text. It accepts the text typed by the user and sends it over the console connection to the switch. Similarly, any bits coming into the PC over the console connection are displayed as text for the user to read.

The emulator must be configured to use the PC's serial port to match the settings on the switch's console port settings. The default console port settings on a switch are as follows. Note that the last three parameters are referred to collectively as 8N1:

Key Topic

- 9600 bits/second
- No hardware flow control
- 8-bit ASCII
- No parity bits
- 1 stop bit

Figure 4-5 shows one such terminal emulator. The image shows the window created by the emulator software in the background, with some output of a **show** command. The foreground, in the upper right, shows a settings window that lists the default console settings as listed just before this paragraph.

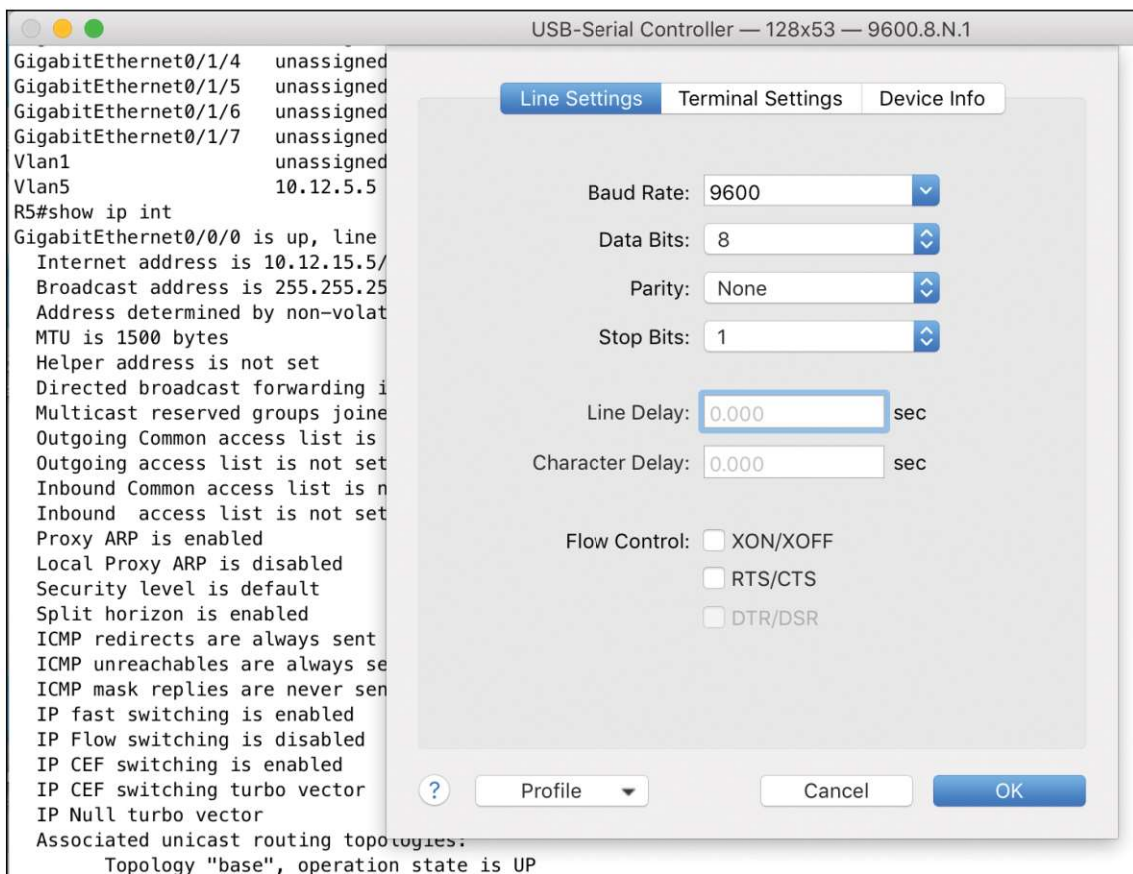


Figure 4-5 Terminal Settings for Console Access

Accessing the CLI with Telnet and SSH

For many years, terminal emulator applications have supported far more than the ability to communicate over a serial port to a local device (like a switch's console). Terminal emulators support a variety of TCP/IP applications as well, including Telnet and SSH. Telnet and SSH both allow the user to connect to another device's CLI, but instead of connecting through

a console cable to the console port, the traffic flows over the same IP network that the net-working devices are helping to create.

Telnet uses the concept of a Telnet client (the terminal application) and a Telnet server (the switch in this case). A *Telnet client*, the device that sits in front of the user, accepts keyboard input and sends those commands to the *Telnet server*. The Telnet server accepts the text, interprets the text as a command, and replies back.

Cisco Catalyst switches enable a Telnet server by default, but switches need a few more con-figuration settings before you can successfully use Telnet to connect to a switch. Chapter 6, “Configuring Basic Switch Management,” covers switch configuration to support Telnet and SSH in detail.

Using Telnet in a lab today makes sense, but Telnet poses a significant security risk in pro-duction networks. Telnet sends all data (including any username and password for login to the switch) as clear-text data. SSH gives us a much better option.

Think of SSH as the much more secure Telnet cousin. Outwardly, you still open a terminal emulator, connect to the switch’s IP address, and see the switch CLI, no matter whether you use Telnet or SSH. The differences exist behind the scenes: SSH encrypts the contents of all messages, including the passwords, avoiding the possibility of someone capturing packets in the network and stealing the password to network devices.

User and Enable (Privileged) Modes

All three CLI access methods covered so far (console, Telnet, and SSH) place the user in an area of the CLI called *user EXEC mode*. User EXEC mode, sometimes also called *user mode*, allows the user to look around but not break anything. The “EXEC mode” part of the name refers to the fact that in this mode, when you enter a command, the switch executes the command and then displays messages that describe the command’s results.

NOTE If you have not used the CLI before, you might want to experiment with the CLI from the Sim Lite product, or view the video about CLI basics. You can find these resources on the companion website as mentioned in the Introduction.

Cisco IOS supports a more powerful EXEC mode called *enable mode* (also known as *privi-leged mode* or *privileged EXEC mode*). Enable mode gets its name from the **enable** com-mand, which moves the user from user mode to enable mode, as shown in Figure 4-6. The other name for this mode, *privileged mode*, refers to the fact that powerful (or privileged) commands can be executed there. For example, you can use the **reload** command, which tells the switch to reinitialize or reboot Cisco IOS, only from enable mode.

NOTE If the command prompt lists the hostname followed by a **>**, the user is in user mode; if it is the hostname followed by the **#**, the user is in enable mode.

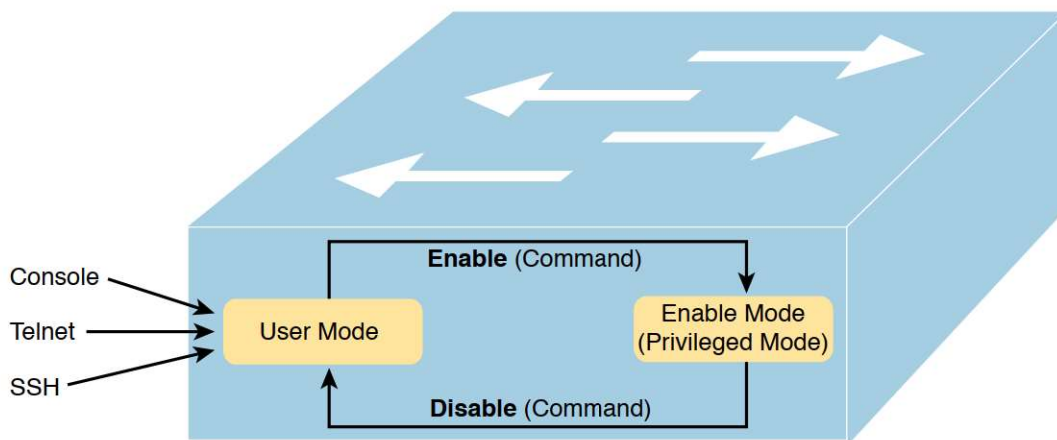


Figure 4-6 *User and Privileged Modes*

Example 4-1 demonstrates the differences between user and enable modes. The example shows the output that you could see in a terminal emulator window, for instance, when connecting from the console. In this case, the user sits at the user mode prompt (“Certskills1>”) and tries the **reload** command. The **reload** command tells the switch to reinitialize or reboot Cisco IOS, so IOS allows this powerful command to be used only from enable mode. IOS rejects the **reload** command when used in user mode. Then the user moves to enable mode—also called privileged mode—(using the **enable EXEC** command). At that point, IOS accepts the **reload** command now that the user is in enable mode.

Example 4-1 *Example of Privileged Mode Commands Being Rejected in User Mode*

```
Press RETURN to get started.

User Access Verification

Password:
Certskills1>
Certskills1> reload
Translating "reload"
% Unknown command or computer name, or unable to find computer address
Certskills1> enable
Password:
Certskills1#
Certskills1# reload

Proceed with reload? [confirm] y
00:08:42: %SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Command.
```

NOTE The commands that can be used in either user (EXEC) mode or enable (EXEC) mode are called EXEC commands.

This example is the first instance of this book showing you the output from the CLI, so it is worth noting a few conventions. The bold text represents what the user typed, and the non-bold text is what the switch sent back to the terminal emulator. Also, the typed passwords do not show up on the screen for security purposes. Finally, note that this switch has been preconfigured with a hostname of Certskills1, so the command prompt on the left shows that hostname on each line.

Password Security for CLI Access from the Console

A Cisco switch, with default settings, remains relatively secure when locked inside a wiring closet, because by default, a switch allows console access only. By default, the console requires no password at all, and no password to reach enable mode for users that happened to connect from the console. The reason is that if you have access to the physical console port of the switch, you already have pretty much complete control over the switch. You could literally get out your screwdriver and walk off with it, or you could unplug the power, or follow well-published procedures to go through password recovery to break into the CLI and then configure anything you want to configure.

However, many people go ahead and set up simple password protection for console users. Simple passwords can be configured at two points in the login process from the console: when the user connects from the console, and when any user moves to enable mode (using the **enable** EXEC command). You may have noticed that back in Example 4-1, the user saw a password prompt at both points.

Example 4-2 shows the additional configuration commands that were configured prior to collecting the output in Example 4-1. The output holds an excerpt from the EXEC command **show running-config**, which lists the current configuration in the switch.

Example 4-2 Nondefault Basic Configuration

```
Certskills1# show running-config
! Output has been formatted to show only the parts relevant to this discussion
hostname Certskills1
!
enable secret love
!
line console 0
  login
  password faith
! The rest of the output has been omitted
Certskills1#
```

Working from top to bottom, note that the first configuration command listed by the **show running-config** command sets the switch's hostname to Certskills1. You might have noticed that the command prompts in Example 4-1 all began with Certskills1, and that's why the command prompt begins with the hostname of the switch.

Next, note that the lines with a ! in them are comment lines, both in the text of this book and in the real switch CLI.

The **enable secret love** configuration command defines the password that all users must use to reach enable mode. So, no matter whether users connect from the console, Telnet, or SSH, they would use the password love when prompted for a password after typing the **enable EXEC** command.

Finally, the last three lines configure the console password. The first line (**line console 0**) is the command that identifies the console, basically meaning “these next commands apply to the console only.” The **login** command tells IOS to perform simple password checking (at the console). Remember, by default, the switch does not ask for a password for console users. Finally, the **password faith** command defines the password the console user must type when prompted.

This example just scratches the surface of the kinds of security configuration you might choose to configure on a switch, but it does give you enough detail to configure switches in your lab and get started (which is the reason I put these details in this first chapter of Part II). Note that Chapter 6 shows the configuration steps to add support for Telnet and SSH (including password security), and Chapter 5 of the *CCNA 200-301 Official Cert Guide, Volume 2*, “Securing Network Devices,” shows additional security configuration as well.

CLI Help Features

If you printed the Cisco IOS Command Reference documents, you would end up with a stack of paper several feet tall. No one should expect to memorize all the commands—and no one does. You can use several very easy, convenient tools to help remember commands and save time typing. As you progress through your Cisco certifications, the exams will cover progressively more commands. However, you should know the methods of getting command help.

Table 4-2 summarizes command-recall help options available at the CLI. Note that, in the first column, *command* represents any command. Likewise, *parm* represents a command’s parameter. For example, the second row lists *command ?*, which means that commands such as **show ?** and **copy ?** would list help for the **show** and **copy** commands, respectively.

Table 4-2 Cisco IOS Software Command Help

What You Enter	What Help You Get
?	Provides help for all commands available in this mode.
<i>command ?</i>	With a space between the command and the ? , the switch lists text to describe all the first parameter options for the command.
<i>com?</i>	Lists commands that start with com .
<i>command parm?</i>	Lists all parameters beginning with the parameter typed so far . (Notice that there is no space between <i>parm</i> and the ? .)
<i>command parm<Tab></i>	Pressing the Tab key causes IOS to spell out the rest of the word, assuming that you have typed enough of the word so there is only one option that begins with that string of characters.
<i>command parm1 ?</i>	If a space is inserted before the question mark, the CLI lists all the next parameters and gives a brief explanation of each.

When you enter the `?`, the Cisco IOS CLI reacts immediately; that is, you don't need to press the Enter key or any other keys. The device running Cisco IOS also redisplay what you entered before the `?` to save you some keystrokes. If you press Enter immediately after the `?`, Cisco IOS tries to execute the command with only the parameters you have entered so far.

The information supplied by using help depends on the CLI mode. For example, when `?` is entered in user mode, the commands allowed in user mode are displayed, but commands available only in enable mode (not in user mode) are not displayed. Also, help is available in configuration mode, which is the mode used to configure the switch. In fact, configuration mode has many different subconfiguration modes, as explained in the section “Configuration Submodes and Contexts,” later in this chapter. So, you can get help for the commands available in each configuration submode as well. (Note that this might be a good time to use the free Sim Lite product on the companion website—open any lab, use the question mark, and try some commands.)

Cisco IOS stores the commands that you enter in a history buffer, storing ten commands by default. The CLI allows you to move backward and forward in the historical list of commands and then edit the command before reissuing it. These key sequences can help you use the CLI more quickly on the exams. Table 4-3 lists the commands used to manipulate previously entered commands.

Table 4-3 Key Sequences for Command Edit and Recall

Keyboard Command	What Happens
Up arrow or Ctrl+P	This displays the most recently used command. If you press it again, the next most recent command appears, until the history buffer is exhausted. (The <i>P</i> stands for previous.)
Down arrow or Ctrl+N	If you have gone too far back into the history buffer, these keys take you forward to the more recently entered commands. (The <i>N</i> stands for next.)
Left arrow or Ctrl+B	This moves the cursor backward in the currently displayed command without deleting characters. (The <i>B</i> stands for back.)
Right arrow or Ctrl+F	This moves the cursor forward in the currently displayed command without deleting characters. (The <i>F</i> stands for forward.)
Backspace	This moves the cursor backward in the currently displayed command, deleting characters.

The debug and show Commands

By far, the single most popular Cisco IOS command is the **show** command. The **show** command has a large variety of options, and with those options, you can find the status of almost every feature of Cisco IOS. Essentially, the **show** command lists the currently known facts about the switch's operational status. The only work the switch does in reaction to **show** commands is to find the current status and list the information in messages sent to the user.

For example, consider the output from the **show mac address-table dynamic** command listed in Example 4-3. This **show** command, issued from user mode, lists the table the switch uses to make forwarding decisions. A switch's MAC address table basically lists the data a switch uses to do its primary job.

Example 4-3 *Nondefault Basic Configuration*

```
Certskills1> show mac address-table dynamic
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
31      0200.1111.1111   DYNAMIC   Gi0/1
31      0200.3333.3333   DYNAMIC   Fa0/3
31      1833.9d7b.0e9a   DYNAMIC   Gi0/1
10      1833.9d7b.0e9a   DYNAMIC   Gi0/1
10      30f7.0d29.8561   DYNAMIC   Gi0/1
1       1833.9d7b.0e9a   DYNAMIC   Gi0/1
12      1833.9d7b.0e9a   DYNAMIC   Gi0/1
Total Mac Addresses for this criterion: 7
Certskills1>
```

The **debug** command also tells the user details about the operation of the switch. However, while the **show** command lists status information at one instant of time—more like a photograph—the **debug** command acts more like a live video camera feed. Once you issue a **debug** command, IOS remembers, issuing messages that any switch user can choose to see. The console sees these messages by default. Most of the commands used throughout this book to verify operation of switches and routers are **show** commands.

Configuring Cisco IOS Software

You will want to configure every switch in an Enterprise network, even though the switches will forward traffic even with default configuration. This section covers the basic configuration processes, including the concept of a configuration file and the locations in which the configuration files can be stored. Although this section focuses on the configuration process, and not on the configuration commands themselves, you should know all the commands covered in this chapter for the exams, in addition to the configuration processes.

Configuration mode is another mode for the Cisco CLI, similar to user mode and privileged mode. User mode lets you issue nondisruptive commands and displays some information. Privileged mode supports a superset of commands compared to user mode, including commands that might disrupt switch operations. However, not one of the commands in user or privileged mode changes the switch's configuration. Configuration mode accepts *configuration commands*—commands that tell the switch the details of what to do and how to do it. Figure 4-7 illustrates the relationships among configuration mode, user EXEC mode, and privileged EXEC mode.

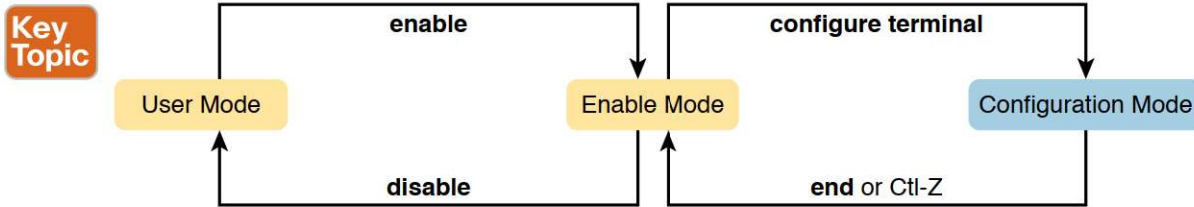


Figure 4-7 CLI Configuration Mode Versus EXEC Modes

Commands entered in configuration mode update the active configuration file. *These changes to the configuration occur immediately each time you press the Enter key at the end of a command.* Be careful when you enter a configuration command!

Configuration Submodes and Contexts

Configuration mode itself contains a multitude of commands. To help organize the configuration, IOS groups some kinds of configuration commands together. To do that, when using configuration mode, you move from the initial mode—global configuration mode—into subcommand modes. *Context-setting commands* move you from one configuration subcommand mode, or context, to another. These context-setting commands tell the switch the topic about which you will enter the next few configuration commands. More importantly, the context tells the switch the topic you care about right now, so when you use the ? to get help, the switch gives you help about that topic only.

NOTE *Context-setting* is not a Cisco term. It is just a description used here to help make sense of configuration mode.

The best way to learn about configuration submodes is to use them, but first, take a look at these upcoming examples. For instance, the **interface** command is one of the most commonly used context-setting configuration commands. For example, the CLI user could enter interface configuration mode by entering the **interface FastEthernet 0/1** configuration command. Asking for help in interface configuration mode displays only commands that are useful when configuring Ethernet interfaces. Commands used in this context are called *subcommands*—or, in this specific case, *interface subcommands*. When you begin practicing with the CLI with real equipment, the navigation between modes can become natural. For now, consider Example 4-4, which shows the following:

- Movement from enable mode to global configuration mode by using the **configure terminal** EXEC command
- Using a **hostname Fred** global configuration command to configure the switch's name
- Movement from global configuration mode to console line configuration mode (using the **line console 0** command)
- Setting the console's simple password to **hope** (using the **password hope** line subcommand)
- Movement from console configuration mode to interface configuration mode (using the **interface type number** command)

- Setting the speed to 100 Mbps for interface Fa0/1 (using the **speed 100** interface subcommand)
- Movement from interface configuration mode back to global configuration mode (using the **exit** command)

Example 4-4 *Navigating Between Different Configuration Modes*

```
Switch# configure terminal
Switch(config)# hostname Fred
Fred(config)# line console 0
Fred(config-line)# password hope
Fred(config-line)# interface FastEthernet 0/1
Fred(config-if)# speed 100
Fred(config-if)# exit
Fred(config)#
```

The text inside parentheses in the command prompt identifies the configuration mode. For example, the first command prompt after you enter configuration mode lists (config), meaning global configuration mode. After the **line console 0** command, the text expands to (config-line), meaning line configuration mode. Each time the command prompt changes within config mode, you have moved to another configuration mode.

Table 4-4 shows the most common command prompts in configuration mode, the names of those modes, and the context-setting commands used to reach those modes.

**Key
Topic**

Table 4-4 Common Switch Configuration Modes

Prompt	Name of Mode	Context-Setting Command(s) to Reach This Mode
hostname(config)#	Global	None—first mode after configure terminal
hostname(config-line)#	Line	line console 0 line vty 0 15
hostname(config-if)#	Interface	interface type number
hostname(vlan)#	VLAN	vlan number

You should practice until you become comfortable moving between the different configuration modes, back to enable mode, and then back into the configuration modes. However, you can learn these skills just doing labs about the topics in later chapters of the book. For now, Figure 4-8 shows most of the navigation between global configuration mode and the four configuration submodes listed in Table 4-4.

NOTE You can also move directly from one configuration submode to another, without first using the **exit** command to move back to global configuration mode. Just use the commands listed in bold in the center of the figure.

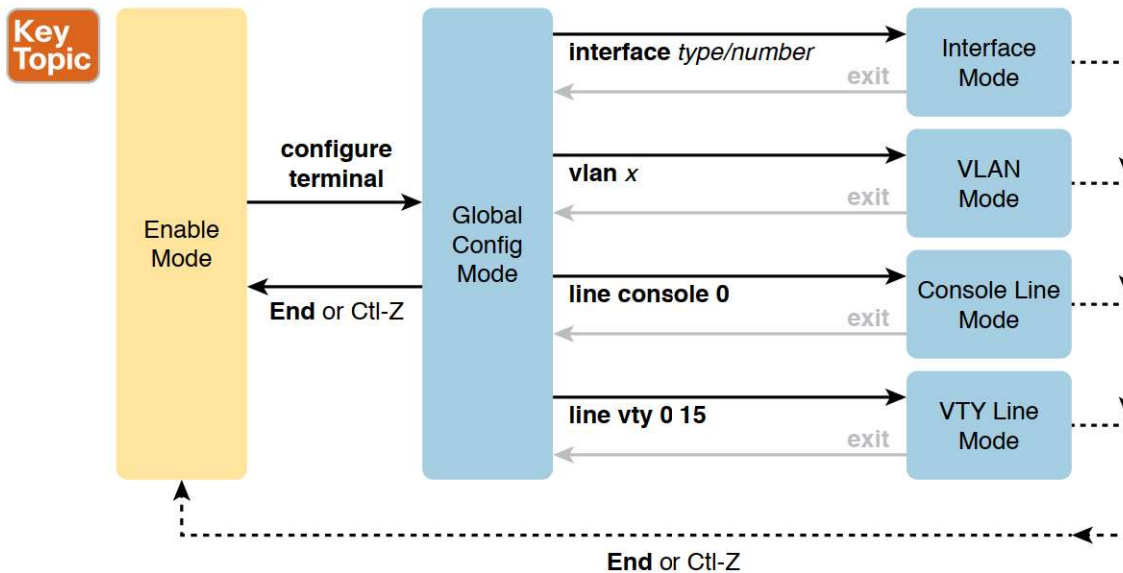


Figure 4-8 *Navigation In and Out of Switch Configuration Modes*

You really should stop and try navigating around these configuration modes. If you have not yet decided on a lab strategy, install the Pearson Sim Lite software from the companion web-site. It includes the simulator and a couple of lab exercises. Start any lab, ignore the instructions, and just get into configuration mode and move around between the configuration modes shown in Figure 4-8.

No set rules exist for what commands are global commands or subcommands. Generally, however, when multiple instances of a parameter can be set in a single switch, the command used to set the parameter is likely a configuration subcommand. Items that are set once for the entire switch are likely global commands. For example, the **hostname** command is a global command because there is only one hostname per switch. Conversely, the **speed** command is an interface subcommand that applies to each switch interface that can run at different speeds, so it is a subcommand, applying to the particular interface under which it is configured.

Storing Switch Configuration Files

When you configure a switch, it needs to use the configuration. It also needs to be able to retain the configuration in case the switch loses power. Cisco switches contain random-access memory (RAM) to store data while Cisco IOS is using it, but RAM loses its contents when the switch loses power or is reloaded. To store information that must be retained when the switch loses power or is reloaded, Cisco switches use several types of more permanent memory, none of which has any moving parts. By avoiding components with moving parts (such as traditional disk drives), switches can maintain better uptime and availability.

The following list details the four main types of memory found in Cisco switches, as well as the most common use of each type:

- **RAM:** Sometimes called DRAM, for dynamic random-access memory, RAM is used by the switch just as it is used by any other computer: for working storage. The running (active) configuration file is stored here.

- **Flash memory:** Either a chip inside the switch or a removable memory card, flash memory stores fully functional Cisco IOS images and is the default location where the switch gets its Cisco IOS at boot time. Flash memory also can be used to store any other files, including backup copies of configuration files.
- **ROM:** Read-only memory (ROM) stores a bootstrap (or boothelper) program that is loaded when the switch first powers on. This bootstrap program then finds the full Cisco IOS image and manages the process of loading Cisco IOS into RAM, at which point Cisco IOS takes over operation of the switch.
- **NVRAM:** Nonvolatile RAM (NVRAM) stores the initial or startup configuration file that is used when the switch is first powered on and when the switch is reloaded.

Figure 4-9 summarizes this same information in a briefer and more convenient form for memorization and study.



Figure 4-9 Cisco Switch Memory Types

Cisco IOS stores the collection of configuration commands in a *configuration file*. In fact, switches use multiple configuration files—one file for the initial configuration used when powering on, and another configuration file for the active, currently used running configuration as stored in RAM. Table 4-5 lists the names of these two files, their purpose, and their storage location.

**Key
Topic**

Table 4-5 Names and Purposes of the Two Main Cisco IOS Configuration Files

Configuration Filename	Purpose	Where It Is Stored
startup-config	Stores the initial configuration used anytime the switch reloads Cisco IOS.	NVRAM
running-config	Stores the currently used configuration commands. This file changes dynamically when someone enters commands in configuration mode.	RAM

Essentially, when you use configuration mode, you change only the running-config file. This means that the configuration example earlier in this chapter (Example 4-4) updates only the running-config file. However, if the switch lost power right after that example, all that configuration would be lost. If you want to keep that configuration, you have to copy the running-config file into NVRAM, overwriting the old startup-config file.

Example 4-5 demonstrates that commands used in configuration mode change only the running configuration in RAM. The example shows the following concepts and steps:

- Step 1.** The example begins with both the running and startup-config having the same hostname, per the **hostname hannah** command.

- Step 2.** The hostname is changed in configuration mode using the **hostname harold** command.
- Step 3.** The **show running-config** and **show startup-config** commands show the fact that the hostnames are now different, with the **hostname harold** command found only in the running-config.

Example 4-5 *How Configuration Mode Commands Change the Running-Config File, Not the Startup-Config File*

```
! Step 1 next (two commands)
!
hannah# show running-config
! (lines omitted)
hostname hannah
! (rest of lines omitted)

hannah# show startup-config
! (lines omitted)
hostname hannah
! (rest of lines omitted)
! Step 2 next. Notice that the command prompt changes immediately after
! the hostname command.

hannah# configure terminal
hannah(config)# hostname harold
harold(config)# exit
! Step 3 next (two commands)
!
harold# show running-config
! (lines omitted) - just showing the part with the hostname command
hostname harold
!
harold# show startup-config
! (lines omitted) - just showing the part with the hostname command
hostname hannah
```

Copying and Erasing Configuration Files

The configuration process updates the running-config file, which is lost if the router loses power or is reloaded. Clearly, IOS needs to provide us a way to copy the running configuration so that it will not be lost, so it will be used the next time the switch reloads or powers on. For instance, Example 4-5 ended with a different running configuration (with the **hostname harold** command) versus the startup configuration.

In short, the EXEC command **copy running-config startup-config** backs up the running-config to the startup-config file. This command overwrites the current startup-config file with what is currently in the running-configuration file.

In addition, in the lab, you may want to just get rid of all existing configuration and start over with a clean configuration. To do that, you can erase the startup-config file using three different commands:

```
write erase
erase startup-config
erase nvram:
```

Once the startup-config file is erased, you can reload or power off/on the switch, and it will boot with the now-empty startup configuration.

Note that Cisco IOS does not have a command that erases the contents of the running-config file. To clear out the running-config file, simply erase the startup-config file, and then **reload** the switch, and the running-config will be empty at the end of the process.

NOTE Cisco uses the term *reload* to refer to what most PC operating systems call rebooting or restarting. In each case, it is a re-initialization of the software. The **reload** EXEC command causes a switch to reload.

Chapter Review

One key to doing well on the exams is to perform repetitive spaced review sessions. Review this chapter's material using either the tools in the book or on the book's companion website. Refer to the "Your Study Plan" element section titled "Step 2: Build Your Study Habits Around the Chapter" for more details. Table 4-6 outlines the key review elements and where you can find them. To better track your study progress, record when you completed these activities in the second column.

Table 4-6 Chapter Review Tracking

Review Element	Review Date(s)	Resource Used
Review key topics		Book, website
Review key terms		Book, website
Repeat DIKTA questions		Book, PTP
Review memory tables		Book, website
Review command tables		Book

Review All the Key Topics



Table 4-7 Key Topics for Chapter 4

Key Topic Element	Description	Page Number
Figure 4-2	Three methods to access a switch CLI	87
Figure 4-3	Cabling options for a console connection	88
List	A Cisco switch's default console port settings	90
Figure 4-7	Navigation between user, enable, and global config modes	97
Table 4-4	A list of configuration mode prompts, the name of the configuration mode, and the command used to reach each mode	98
Figure 4-8	Configuration mode context-setting commands	99
Table 4-5	The names and purposes of the two configuration files in a switch or router	100

Key Terms You Should Know

command-line interface (CLI), Telnet, Secure Shell (SSH), enable mode, user mode, configuration mode, startup-config file, running-config file

Command References

Tables 4-8 and 4-9 list configuration and verification commands used in this chapter, respectively. As an easy review exercise, cover the left column in a table, read the right column, and try to recall the command without looking. Then repeat the exercise, covering the right column, and try to recall what the command does.

Table 4-8 Chapter 4 Configuration Commands

Command	Mode and Purpose
line console 0	Global command that changes the context to console configuration mode.
login	Line (console and vty) configuration mode. Tells IOS to prompt for a password (no username).
password <i>pass-value</i>	Line (console and vty) configuration mode. Sets the password required on that line for login if the login command (with no other parameters) is also configured.
interface <i>type port-number</i>	Global command that changes the context to interface mode—for example, interface FastEthernet 0/1 .
hostname <i>name</i>	Global command that sets this switch's hostname, which is also used as the first part of the switch's command prompt.
exit	Moves back to the next higher mode in configuration mode.

Command	Mode and Purpose
end	Exits configuration mode and goes back to enable mode from any of the configuration submodes.
Ctrl+Z	This is not a command, but rather a two-key combination (pressing the Ctrl key and the letter Z) that together do the same thing as the end command.

Table 4-9 Chapter 4 EXEC Command Reference

Command	Purpose
no debug all undebug all	Enable mode EXEC command to disable all currently enabled debugs.
reload	Enable mode EXEC command that reboots the switch or router.
copy running-config startup-config	Enable mode EXEC command that saves the active config, replacing the startup-config file used when the switch initializes.
copy startup-config running-config	Enable mode EXEC command that merges the startup-config file with the currently active config file in RAM.
show running-config	Lists the contents of the running-config file.
write erase erase startup-config erase nvram:	These enable mode EXEC commands erase the startup-config file.
quit	EXEC command that disconnects the user from the CLI session.
show startup-config	Lists the contents of the startup-config (initial config) file.
enable	Moves the user from user mode to enable (privileged) mode and prompts for a password if one is configured.
disable	Moves the user from enable mode to user mode.
configure terminal	Enable mode command that moves the user into configuration mode.

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CHAPTER 5

Analyzing Ethernet LAN Switching

This chapter covers the following exam topics:

1.0 Network Fundamentals

1.1 Explain the role and function of network components

1.1.b L2 and L3 Switches

1.13 Describe switching concepts

1.13.a MAC learning and aging

1.13.b Frame switching

1.13.c Frame flooding

1.13.d MAC address table

2.0 Network Access

2.5 Describe the need for and basic operations of Rapid PVST+ Spanning Tree Protocol and identify basic operations

When you buy a Cisco Catalyst Ethernet switch, the switch is ready to work. All you have to do is take it out of the box, power on the switch by connecting the power cable to the switch and a power outlet, and connect hosts to the switch using the correct unshielded twisted-pair (UTP) cables. You do not have to configure anything else, or connect to the console and login, or do anything; the switch just starts forwarding Ethernet frames.

In Part II of this book, you will learn how to build, configure, and verify the operation of Ethernet LANs. In Chapter 4, “Using the Command-Line Interface,” you learned how to move around in the CLI, issue commands, and configure the switch. This chapter takes a short but important step in that journey by explaining the logic a switch uses when forwarding Ethernet frames.

This chapter breaks the content into two major sections. The first reviews and then further develops the concepts behind LAN switching, which were first introduced back in Chapter 2, “Fundamentals of Ethernet LANs.” The second section then uses IOS show commands to verify that Cisco switches actually learned the MAC addresses, built the MAC address table, and forwarded frames.

“Do I Know This Already?” Quiz

Take the quiz (either here or use the PTP software) if you want to use the score to help you decide how much time to spend on this chapter. The letter answers are listed at the bottom of the page following the quiz. Appendix C, found both at the end of the book as well as on the companion website, includes both the answers and explanations. You can also find both answers and explanations in the PTP testing software.

Table 5-1 “Do I Know This Already?” Foundation Topics Section-to-Question Mapping

Foundation Topics Section	Questions
LAN Switching Concepts	1–4
Verifying and Analyzing Ethernet Switching	5–6

1. Which of the following statements describes part of the process of how a switch decides to forward a frame destined for a known unicast MAC address?
 - a. It compares the unicast destination address to the bridging, or MAC address, table.
 - b. It compares the unicast source address to the bridging, or MAC address, table.
 - c. It forwards the frame out all interfaces in the same VLAN except for the incoming interface.
 - d. It compares the destination IP address to the destination MAC address.
 - e. It compares the frame’s incoming interface to the source MAC entry in the MAC address table.
2. Which of the following statements describes part of the process of how a LAN switch decides to forward a frame destined for a broadcast MAC address?
 - a. It compares the unicast destination address to the bridging, or MAC address, table.
 - b. It compares the unicast source address to the bridging, or MAC address, table.
 - c. It forwards the frame out all interfaces in the same VLAN except for the incoming interface.
 - d. It compares the destination IP address to the destination MAC address.
 - e. It compares the frame’s incoming interface to the source MAC entry in the MAC address table.
3. Which of the following statements best describes what a switch does with a frame destined for an unknown unicast address?
 - a. It forwards out all interfaces in the same VLAN except for the incoming interface.
 - b. It forwards the frame out the one interface identified by the matching entry in the MAC address table.
 - c. It compares the destination IP address to the destination MAC address.
 - d. It compares the frame’s incoming interface to the source MAC entry in the MAC address table.
4. Which of the following comparisons does a switch make when deciding whether a new MAC address should be added to its MAC address table?
 - a. It compares the unicast destination address to the bridging, or MAC address, table.
 - b. It compares the unicast source address to the bridging, or MAC address, table.
 - c. It compares the VLAN ID to the bridging, or MAC address, table.
 - d. It compares the destination IP address’s ARP cache entry to the bridging, or MAC address, table.

5. A Cisco Catalyst switch has 24 10/100 ports, numbered 0/1 through 0/24. Ten PCs connect to the 10 lowest numbered ports, with those PCs working and sending data over the network. The other ports are not connected to any device. Which of the following answers lists facts displayed by the show interfaces status command?
- Port Ethernet 0/1 is in a connected state.
 - Port Fast Ethernet 0/11 is in a connected state.
 - Port Fast Ethernet 0/5 is in a connected state.
 - Port Ethernet 0/15 is in a notconnected state.

6. Consider the following output from a Cisco Catalyst switch:

```
SW1# show mac address-table dynamic
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       02AA.AAAA.AAAA   DYNAMIC   Gi0/1
1       02BB.BBBB.BBBB   DYNAMIC   Gi0/2
1       02CC.CCCC.CCCC   DYNAMIC   Gi0/3
Total Mac Addresses for this criterion: 3
```

Which of the following answers is true about this switch?

- The output proves that port Gi0/2 connects directly to a device that uses address 02BB.BBBB.BBBB.
- The switch has learned three MAC addresses since the switch powered on.
- The three listed MAC addresses were learned based on the destination MAC address of frames forwarded by the switch.
- 02CC.CCCC.CCCC was learned from the source MAC address of a frame that entered port Gi0/3.

Foundation Topics

LAN Switching Concepts

A modern Ethernet LAN connects user devices as well as servers into some switches, with the switches then connecting to each other, sometimes in a design like Figure 5-1. Part of the LAN, called a campus LAN, supports the end-user population as shown on the left of the figure. End-user devices connect to LAN switches, which in turn connect to other switches so that a path exists to the rest of the network. The campus LAN switches sit in wiring closets close to the end users. On the right, the servers used to provide information to the users also connect to the LAN. Those servers and switches often sit in a closed room called a *data center*, with connections to the campus LAN to support traffic to/from the users.

To forward traffic from a user device to a server and back, each switch performs the same kind of logic, independently from each other. The first half of this chapter examines the logic: how a switch chooses to forward an Ethernet frame, when the switch chooses to not forward the frame, and so on.

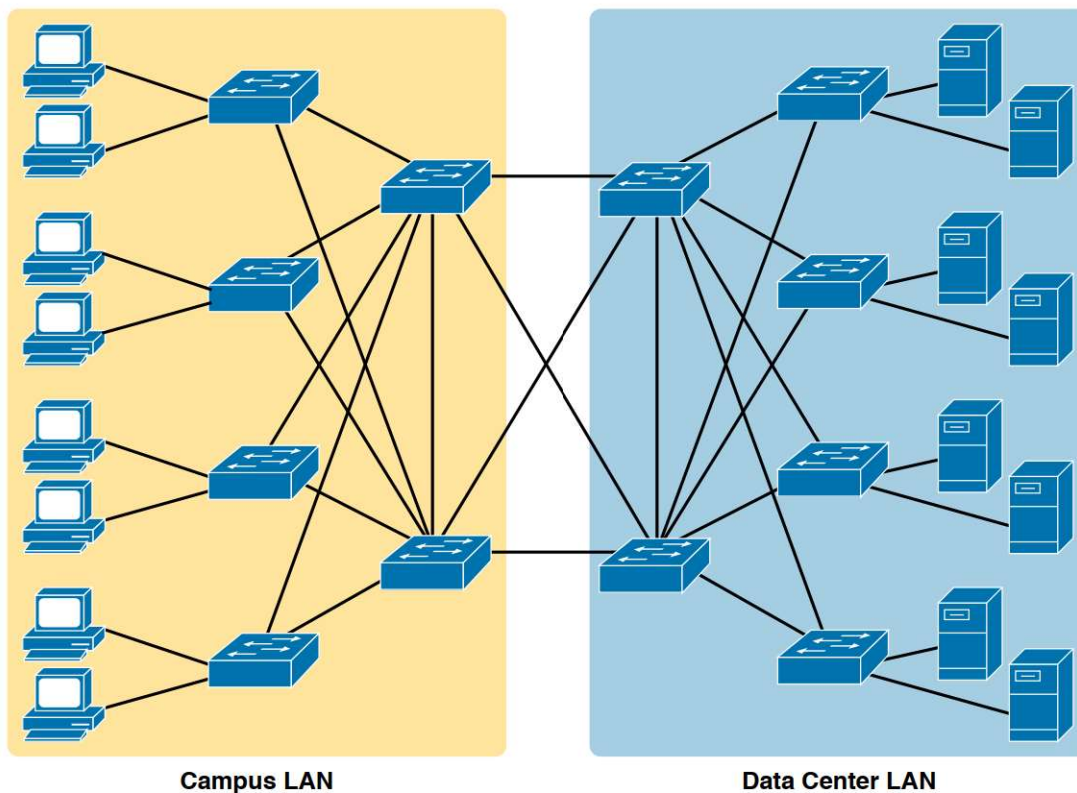


Figure 5-1 *Campus LAN and Data Center LAN, Conceptual Drawing*

Overview of Switching Logic

Ultimately, the role of a LAN switch is to forward Ethernet frames. LANs exist as a set of user devices, servers, and other devices that connect to switches, with the switches connected to each other. The LAN switch has one primary job: to forward frames to the correct destination (MAC) address. And to achieve that goal, switches use logic—logic based on the source and destination MAC address in each frame's Ethernet header.

LAN switches receive Ethernet frames and then make a switching decision: either forward the frame out some other ports or ignore the frame. To accomplish this primary mission, switches perform three actions:

Key Topic

1. Deciding when to forward a frame or when to filter (not forward) a frame, based on the destination MAC address
2. Preparing to forward frames by learning MAC addresses by examining the source MAC address of each frame received by the switch
3. Preparing to forward only one copy of the frame to the destination by creating a (Layer 2) loop-free environment with other switches by using Spanning Tree Protocol (STP)

The first action is the switch's primary job, whereas the other two items are overhead functions.

NOTE Throughout this book's discussion of LAN switches, the terms *switch port* and *switch interface* are synonymous.

Although Chapter 2's section titled "Ethernet Data-Link Protocols" already discussed the frame format, this discussion of Ethernet switching is pretty important, so reviewing the Ethernet frame at this point might be helpful. Figure 5-2 shows one popular format for an Ethernet frame. Basically, a switch would take the frame shown in the figure, make a decision of where to forward the frame, and send the frame out that other interface.

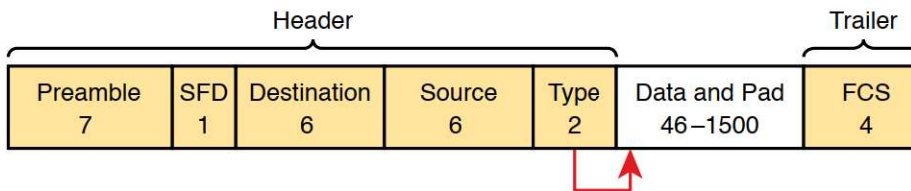


Figure 5-2 IEEE 802.3 Ethernet Frame (One Variation)

Most of the upcoming discussions and figures about Ethernet switching focus on the use of the destination and source MAC address fields in the header. All Ethernet frames have both a destination and source MAC address. Both are 6-bytes long (represented as 12 hex digits in the book) and are a key part of the switching logic discussed in this section. Refer back to Chapter 2's discussion of the header in detail for more info on the rest of the Ethernet frame.

NOTE The companion website includes a video that explains the basics of Ethernet switching.

Now on to the details of how Ethernet switching works!

Forwarding Known Unicast Frames

To decide whether to forward a frame, a switch uses a dynamically built table that lists MAC addresses and outgoing interfaces. Switches compare the frame's destination MAC address to this table to decide whether the switch should forward a frame or simply ignore it. For example, consider the simple network shown in Figure 5-3, with Fred sending a frame to Barney.

In this figure, Fred sends a frame with destination address 0200.2222.2222 (Barney's MAC address). The switch compares the destination MAC address (0200.2222.2222) to the MAC address table, matching the bold table entry. That matched table entry tells the switch to forward the frame out port F0/2, and only port F0/2.

Answers to the "Do I Know This Already?" quiz:

1 A 2 C 3 A 4 B 5 C 6 D

NOTE A switch's MAC address table is also called the *switching table*, or *bridging table*, or even the *Content-Addressable Memory (CAM) table*, in reference to the type of physical memory used to store the table.

Key Topic

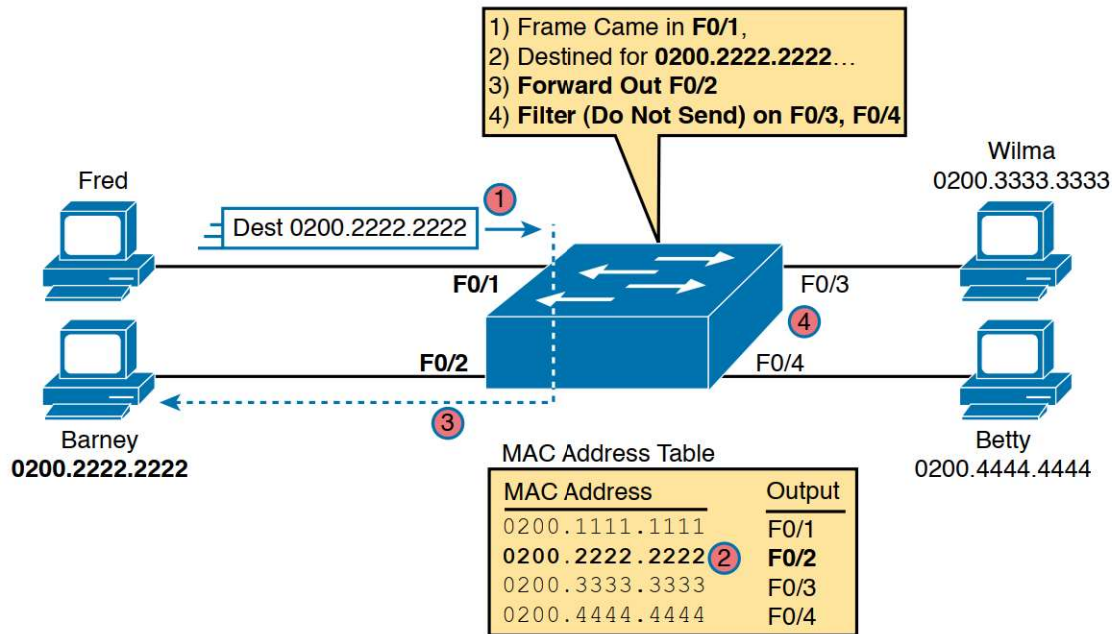


Figure 5-3 Sample Switch Forwarding and Filtering Decision

A switch's MAC address table lists the location of each MAC relative to that one switch. In LANs with multiple switches, each switch makes an independent forwarding decision based on its own MAC address table. Together, they forward the frame so that it eventually arrives at the destination.

For example, Figure 5-4 shows the first switching decision in a case in which Fred sends a frame to Wilma, with destination MAC 0200.3333.3333. The topology has changed versus the previous figure, this time with two switches, and Fred and Wilma connected to two different switches. Figure 5-3 shows the first switch's logic, in reaction to Fred sending the original frame. Basically, the switch receives the frame in port F0/1, finds the destination MAC (0200.3333.3333) in the MAC address table, sees the outgoing port of G0/1, so SW1 forwards the frame out its G0/1 port.

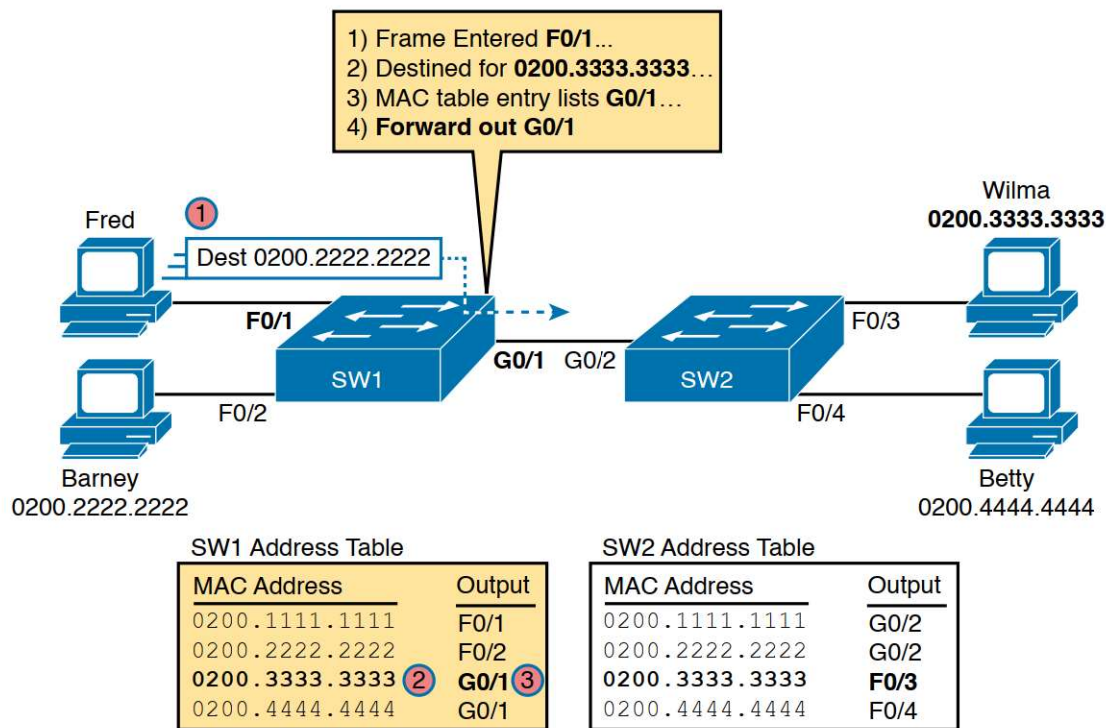


Figure 5-4 Forwarding Decision with Two Switches: First Switch

That same frame next arrives at switch SW2, entering SW2's G0/2 interface. As shown in Figure 5-5, SW2 uses the same logic steps, but using SW2's table. The MAC table lists the forwarding instructions for that switch only. In this case, switch SW2 forwards the frame out its F0/3 port, based on SW2's MAC address table.

Key Topic

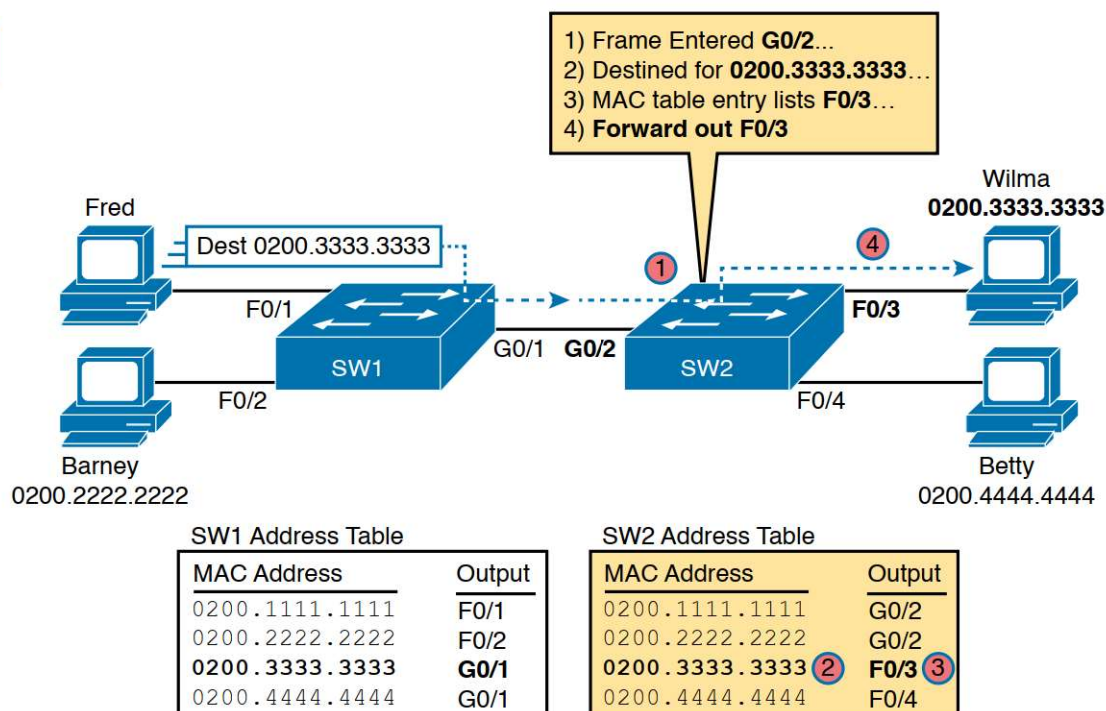


Figure 5-5 Forwarding Decision with Two Switches: Second Switch

NOTE The forwarding choice by a switch was formerly called a *forward-versus-filter* decision, because the switch also chooses to not forward (to filter) frames, not sending the frame out some ports.

The examples so far use switches that happen to have a MAC table with all the MAC addresses listed. As a result, the destination MAC address in the frame is known to the switch. The frames are called known unicast frames, or simply known unicasts, because the destination address is a unicast address, and the destination is known. As shown in these examples, switches forward known unicast frames out one port: the port as listed in the MAC table entry for that MAC address.

Learning MAC Addresses

Thankfully, the networking staff does not have to type in all those MAC table entries. Instead, each switch does its second main function: to learn the MAC addresses and interfaces to put into its address table. With a complete MAC address table, the switch can make accurate forwarding and filtering decisions as just discussed.

Switches build the address table by listening to incoming frames and examining the *source MAC address* in the frame. If a frame enters the switch and the source MAC address is not in the MAC address table, the switch creates an entry in the table. That table entry lists the interface from which the frame arrived. Switch learning logic is that simple.

Figure 5-6 depicts the same single-switch topology network as Figure 5-3, but before the switch has built any address table entries. The figure shows the first two frames sent in this network—first a frame from Fred, addressed to Barney, and then Barney’s response, addressed to Fred.

5

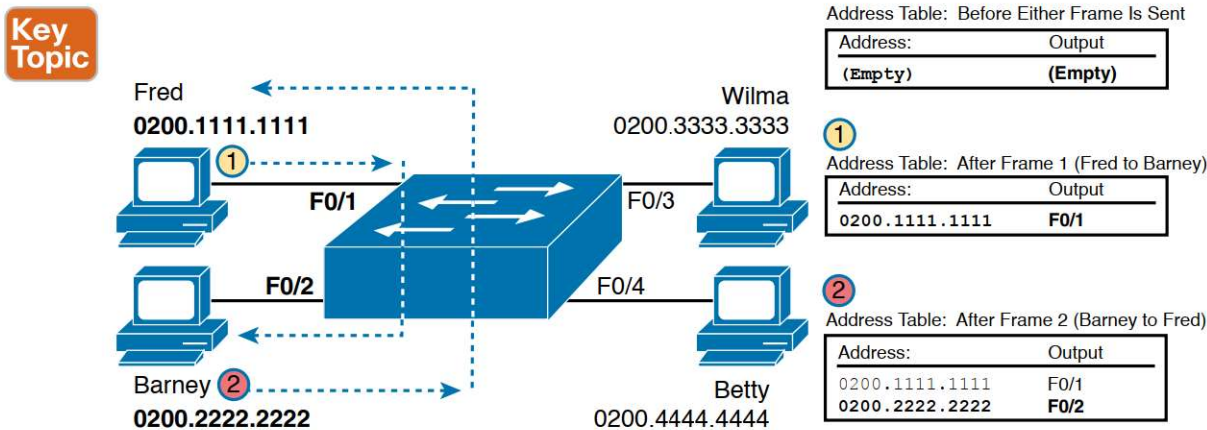


Figure 5-6 Switch Learning: Empty Table and Adding Two Entries

(Figure 5-6 depicts the MAC learning process only, and ignores the forwarding process and therefore ignores the destination MAC addresses.)

Focus on the learning process and how the MAC table grows at each step as shown on the right side of the figure. The switch begins with an empty MAC table, as shown in the upper-right part of the figure. Then Fred sends his first frame (labeled “1”) to Barney, so the switch

adds an entry for 0200.1111.1111, Fred's MAC address, associated with interface F0/1. Why F0/1? The frame sent by Fred entered the switch's F0/1 port. SW1's logic runs something like this: "The source is MAC 0200.1111.1111, the frame entered F0/1, so from my perspective, 0200.1111.1111 must be reachable out my port F0/1."

Continuing the example, when Barney replies in Step 2, the switch adds a second entry, this one for 0200.2222.2222, Barney's MAC address, along with interface F0/2. Why F0/2? The frame Barney sent entered the switch's F0/2 interface. Learning always occurs by looking at the source MAC address in the frame and adds the incoming interface as the associated port.

Flooding Unknown Unicast and Broadcast Frames

Now again turn your attention to the forwarding process, using the topology in Figure 5-5. What do you suppose the switch does with Fred's first frame, the one that occurred when there were no entries in the MAC address table? As it turns out, when there is no matching entry in the table, switches forward the frame out all interfaces (except the incoming interface) using a process called *flooding*. And the frame whose destination address is unknown to the switch is called an *unknown unicast frame*, or simply an *unknown unicast*.

Switches flood unknown unicast frames. Flooding means that the switch forwards copies of the frame out all ports, except the port on which the frame was received. The idea is simple: if you do not know where to send it, send it everywhere, to deliver the frame. And, by the way, that device will likely then send a reply—and then the switch can learn that device's MAC address and forward future frames out one port as a known unicast frame.

Switches also flood LAN broadcast frames (frames destined to the Ethernet broadcast address of FFFF.FFFF.FFFF) because this process helps deliver a copy of the frame to all devices in the LAN.

For example, Figure 5-7 shows the same first frame sent by Fred, when the switch's MAC table is empty. At step 1, Fred sends the frame. At step 2, the switch sends a copy of the frame out all three of the other interfaces.

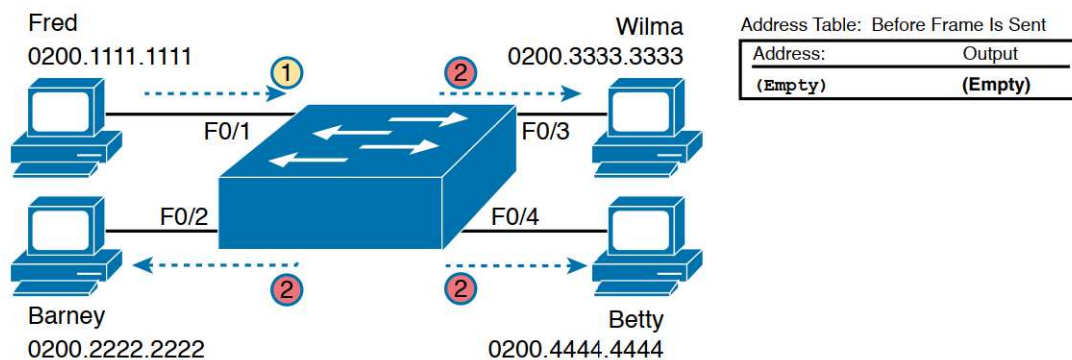


Figure 5-7 Switch Flooding: Unknown Unicast Arrives, Floods Out Other Ports

Avoiding Loops Using Spanning Tree Protocol

The third primary feature of LAN switches is loop prevention, as implemented by Spanning Tree Protocol (STP). Without STP, any flooded frames would loop for an indefinite period of

time in Ethernet networks with physically redundant links. To prevent looping frames, STP blocks some ports from forwarding frames so that only one active path exists between any pair of LAN segments.

The result of STP is good: frames do not loop infinitely, which makes the LAN usable. However, STP has negative features as well, including the fact that it takes some work to balance traffic across the redundant alternate links.

A simple example makes the need for STP more obvious. Remember, switches flood unknown unicast frames and broadcast frames. Figure 5-8 shows an unknown unicast frame, sent by Larry to Bob, which loops forever because the network has redundancy but no STP. Note that the figure shows one direction of the looping frame only, just to reduce clutter, but a copy of the frame would also loop the other direction.

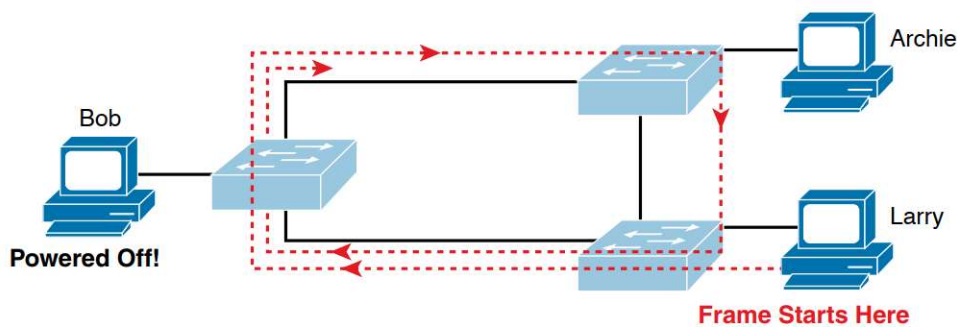


Figure 5-8 Network with Redundant Links but Without STP: The Frame Loops Forever

The flooding of this frame would result in the frame repeatedly rotating around the three switches, because none of the switches list Bob's MAC address in their address tables—so each switch floods the frame. And while the flooding process is a good mechanism for forwarding unknown unicasts and broadcasts, the continual flooding of traffic frames as in the figure can completely congest the LAN to the point of making it unusable.

A topology like Figure 5-8, with redundant links, is good, but we need to prevent the bad effect of those looping frames. To avoid Layer 2 loops, all switches need to use STP. STP causes each interface on a switch to settle into either a blocking state or a forwarding state. *Blocking* means that the interface cannot forward or receive data frames, while *forwarding* means that the interface can send and receive data frames. If a correct subset of the interfaces is blocked, only a single currently active logical path exists between each pair of LANs.

NOTE STP behaves identically for a transparent bridge and a switch. Therefore, the terms *bridge*, *switch*, and *bridging device* all are used interchangeably when discussing STP.

Chapter 9 of this book, “Spanning Tree Protocol Concepts,” examines STP in depth, including how STP prevents loops.

LAN Switching Summary

Switches use Layer 2 logic, examining the Ethernet data-link header to choose how to process frames. In particular, switches make decisions to forward and filter frames, learn MAC addresses, and use STP to avoid loops, as follows:

- Step 1.** Switches forward frames based on the destination MAC address:
- A.** If the destination MAC address is a broadcast, multicast, or unknown destination unicast (a unicast not listed in the MAC table), the switch floods the frame.
 - B.** If the destination MAC address is a known unicast address (a unicast address found in the MAC table):
 - i.** If the outgoing interface listed in the MAC address table is different from the interface in which the frame was received, the switch forwards the frame out the outgoing interface.
 - ii.** If the outgoing interface is the same as the interface in which the frame was received, the switch filters the frame, meaning that the switch simply ignores the frame and does not forward it.
- Step 2.** Switches use the following logic to learn MAC address table entries:
- A.** For each received frame, examine the source MAC address and note the interface from which the frame was received.
 - B.** If it is not already in the table, add the MAC address and interface it was learned on.
- Step 3.** Switches use STP to prevent loops by causing some interfaces to block, meaning that they do not send or receive frames.

Verifying and Analyzing Ethernet Switching

A Cisco Catalyst switch comes from the factory ready to switch frames. All you have to do is connect the power cable, plug in the Ethernet cables, and the switch starts switching incoming frames. Connect multiple switches together, and they are ready to forward frames between the switches as well. And the big reason behind this default behavior has to do with the default settings on the switches.

Cisco Catalyst switches come ready to get busy switching frames because of settings like these:

- The interfaces are enabled by default, ready to start working once a cable is connected.
- All interfaces are assigned to VLAN 1.
- 10/100 and 10/100/1000 interfaces use autonegotiation by default.
- The MAC learning, forwarding, flooding logic all works by default.
- STP is enabled by default.

This second section of the chapter examines how switches will work with these default settings, showing how to verify the Ethernet learning and forwarding process.

Demonstrating MAC Learning

To see a switch's MAC address table, use the **show mac address-table** command. With no additional parameters, this command lists all known MAC addresses in the MAC table, including some overhead static MAC addresses that you can ignore. To see all the dynamically learned MAC addresses only, instead use the **show mac address-table dynamic** command.

The examples in this chapter use almost no configuration, as if you just unboxed the switch when you first purchased it. For the examples, the switches have no configuration other than the **hostname** command to set a meaningful hostname. Note that to do this in lab, all I did was

Key Topic

- Use the **erase startup-config EXEC** command to erase the startup-config file
- Use the **delete vlan.dat EXEC** command to delete the VLAN configuration details
- Use the **reload EXEC** command to reload the switch (thereby using the empty startup-config, with no VLAN information configured)
- Configure the **hostname SW1** command to set the switch hostname

Once done, the switch starts forwarding and learning MAC addresses, as demonstrated in Example 5-1.

5

Key Topic

Example 5-1 show mac address-table dynamic for Figure 5-7

```
SW1# show mac address-table dynamic
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0200.1111.1111   DYNAMIC Fa0/1
1       0200.2222.2222   DYNAMIC Fa0/2
1       0200.3333.3333   DYNAMIC Fa0/3
1       0200.4444.4444   DYNAMIC Fa0/4
Total Mac Addresses for this criterion: 4
SW1#
```

First, focus on two columns of the table: the MAC Address and Ports columns of the table. The values should look familiar: they match the earlier single-switch example, as repeated here as Figure 5-9. Note the four MAC addresses listed, along with their matching ports, as shown in the figure.